# JPWM07 Light refraction — Quiz 2

S3 Physics · 3 multiple-choice items

### Q1 · JPWM07.3 · Foundation

The speed of light in a medium is $2 \times 10^8$ m $s^{-1}$. What is the refractive index of the medium? (Given that the speed of light in vacuum = $3 \times 10^8$ m $s^{-1}$)

- **A.** 0.67
- **B.** 1.33
- **C.** 1.50
- **D.** 2.00

### Q2 · JPWM07.3 · Standard

When a light ray travels from air to water with an angle of incidence of 45°, the angle of refraction is 32.1°. What is the speed of light in the water? (Given that the speed of light in vacuum = $3 \times 10^8$ m $s^{-1}$)

- **A.** $2.14 \times 10^8$ m $s^{-1}$
- **B.** $2.25 \times 10^8$ m $s^{-1}$
- **C.** $2.42 \times 10^8$ m $s^{-1}$
- **D.** $2.59 \times 10^8$ m $s^{-1}$

### Q3 · JPWM07.4 · Standard

A boy is looking at small print on a piece of paper. He moves a glass block of uniform thickness 3 cm over the print. The print appears to rise by 0.9 cm. What is the refractive index of the glass?

- **A.** 0.70
- **B.** 1.30
- **C.** 1.43
- **D.** 3.33

# JPWM07 Light refraction — Quiz 2 · Answer key

Teacher page · 3 items

### Q1 · JPWM07.3

**Answer:** C. 1.50

*Guide:* n = c / v = (3 × $10^8$ m $s^{-1}$) / (2 × $10^8$ m $s^{-1}$) = 1.50. Option A is v / c.

### Q2 · JPWM07.3

**Answer:** B. 2.25 × $10^8$ m $s^{-1}$

*Guide:* By Snell's law, n = sin 45° / sin 32.1° ≈ 1.33. Then n = c / v, so v = c / n = (3 × $10^8$ m $s^{-1}$) / 1.33 ≈ 2.25 × $10^8$ m $s^{-1}$.

### Q3 · JPWM07.4

**Answer:** C. 1.43

*Guide:* Apparent depth = real depth − rise = 3 cm − 0.9 cm = 2.1 cm. n = real depth / apparent depth = 3 / 2.1 ≈ 1.43. Option A is 2.1 / 3. Option D uses the rise instead of the apparent depth (3 / 0.9).